

College Management System Project with SQL Code

Phase 1 – Design

Entities: Students, Professors, Courses, Departments, Hostels, Fees, Results, Enrollments, Hostel_Allotments.

Phase 2 – Create Tables

```
CREATE TABLE Departments(  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50) UNIQUE NOT NULL  
);  
  
CREATE TABLE Students(  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    cgpa DECIMAL(3,2),  
    dept_id INT,  
    FOREIGN KEY(dept_id) REFERENCES Departments(dept_id)  
    ON DELETE CASCADE  
);
```

Phase 3 – Populate Data

```
INSERT INTO Departments VALUES  
(1, 'CSE'), (2, 'ECE'), (3, 'EEE'), (4, 'MECH'), (5, 'CIVIL');  
  
UPDATE Students  
SET cgpa=8.5  
WHERE student_id=101;  
  
DELETE FROM Students  
WHERE student_id=108;
```

Phase 4 – Indexes

```
CREATE INDEX idx_student_name  
ON Students(student_name);  
  
EXPLAIN  
SELECT * FROM Students  
WHERE student_name='Navya';
```

Phase 5 – Queries

```
SELECT s.student_name, d.dept_name  
FROM Students s  
JOIN Departments d  
ON s.dept_id=d.dept_id;  
  
SELECT dept_id, COUNT(*)  
FROM Students  
GROUP BY dept_id;
```

Phase 6 – Transactions

```
START TRANSACTION;  
  
INSERT INTO Enrollments  
VALUES(1, 101, 201);  
  
COMMIT;  
START TRANSACTION;
```

```
INSERT INTO Results VALUES(1,101,201,'A');  
SAVEPOINT result_saved;  
INSERT INTO Results VALUES(2,101,202,'A');  
ROLLBACK TO result_saved;  
COMMIT;
```

Phase 7 – Practice

10 SQL practice tasks, self-assessment checklist, and extension projects.